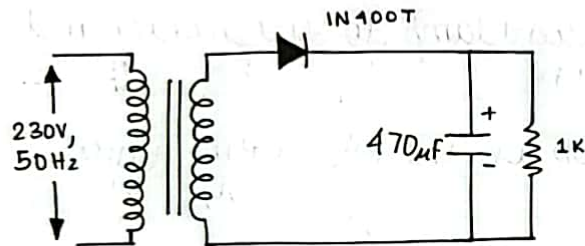
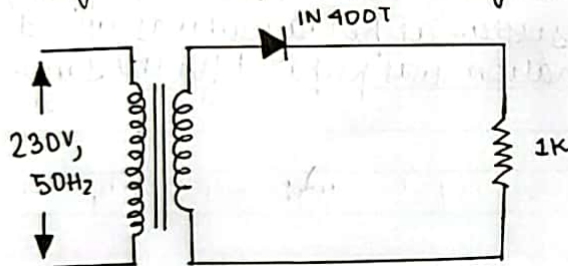


CIRCUIT DIAGRAM

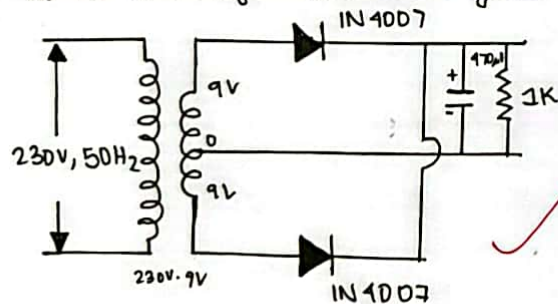
Half wave rectifier circuit diagram (with filter)



Half wave rectifier circuit diagram (without filter)



Full wave rectifier circuit diagram (with filter)



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RECTIFIER CIRCUITS

AIM

To study the performance of half and full wave (centre tapped) rectifier circuits

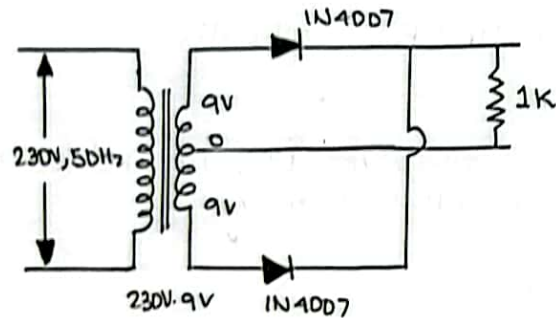
- i) with filter
- ii) without filter

MATERIALS AND EQUIPMENT REQUIRED

S.No	ITEM	SPECIFICATION	QUANTITY
1.	Transformer	(9-0-9)V, 500mA	1
2.	Diode	IN4007	2
3.	Variable Resistor	1K Ω	1
4.	Fixed Resistor	1K Ω	1
5.	Capacitor	470µF, 1000µF	1 each
6.	CRO		
7.	Bread board		

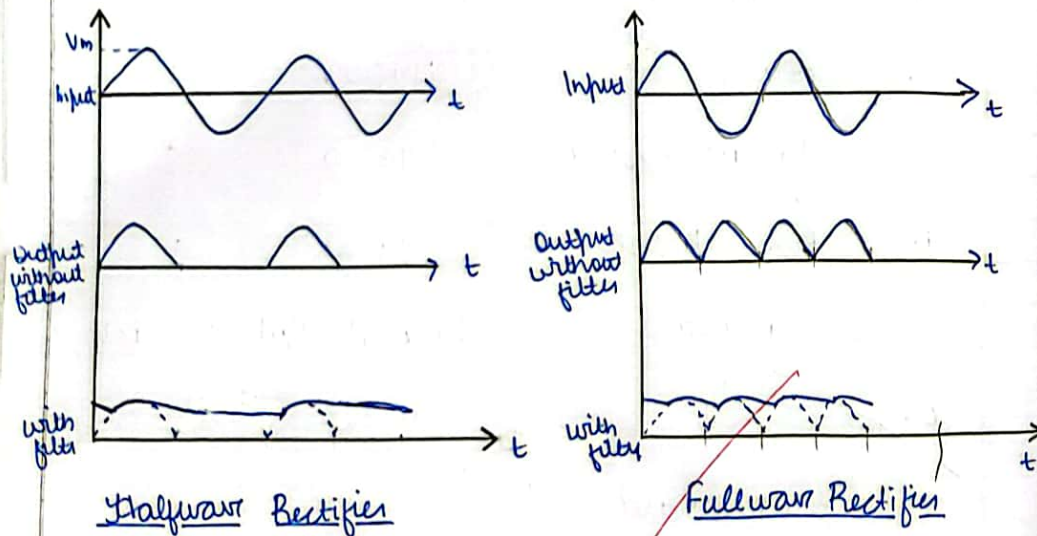
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Full wave rectifier circuit diagram (without filter)



OBSERVATIONS

Wave forms.



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THEORY

Half Wave Rectifier :

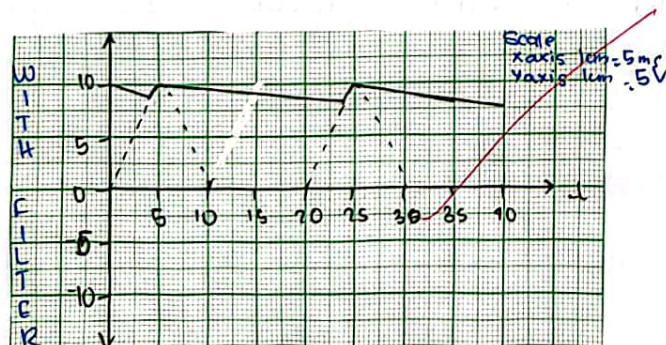
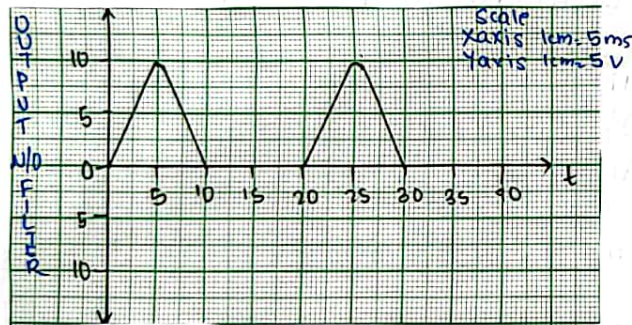
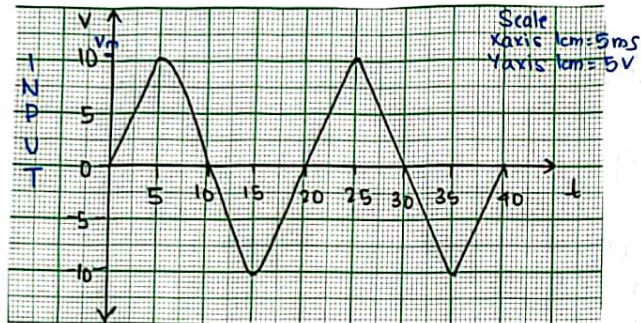
A simple half wave rectifier is nothing more than a single pn junction diode connected in series to a load resistor. Input voltage is given to a step down transformer and the resulting reduced output of transformer is given to the diode 'D' and load resistor R_L . The output voltage is measured across load resistor R_L .

The half wave rectifier circuit using a semiconductor diode (D) with a load resistance R_L but no smoothing filter is given in figure. The ac voltage across the secondary winding changes polarity after every half cycle of input wave. During the positive half cycles of the input ac voltage, during the negative half cycles of the input ac voltage is when the lower end of the secondary winding is positive w.r.t its upper end, the diode is reverse biased and so it does not conduct. Thus during the negative half cycles of the input ac voltage, the current through and voltage across the load remains zero.

Thus the output voltage (V_L) developed across load resistance R_L is a series of positive half cycles of alternating voltage, with intervening very

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HALF WAVE RECTIFIER



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constant negative voltage levels, it is obvious from the figure that the output is not a steady dc, but only a pulsating dc wave. To make the output wave smooth and useful in a DC power supply, we have to use a filter across the load.

Full Wave Rectifier Circuit

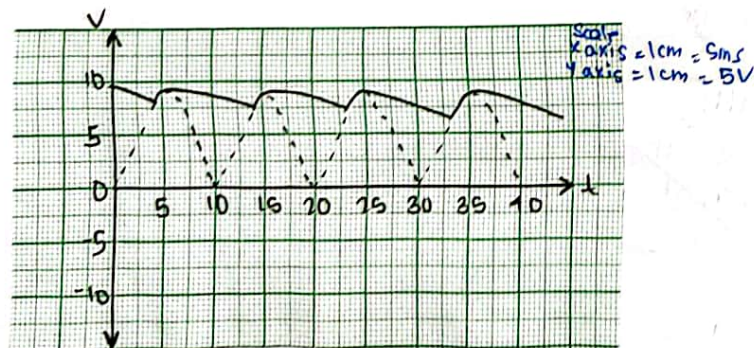
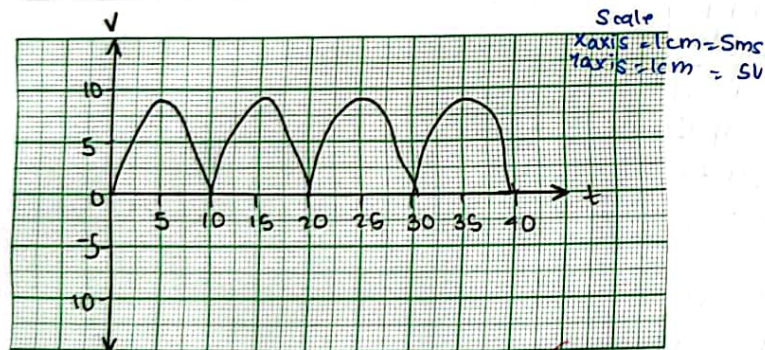
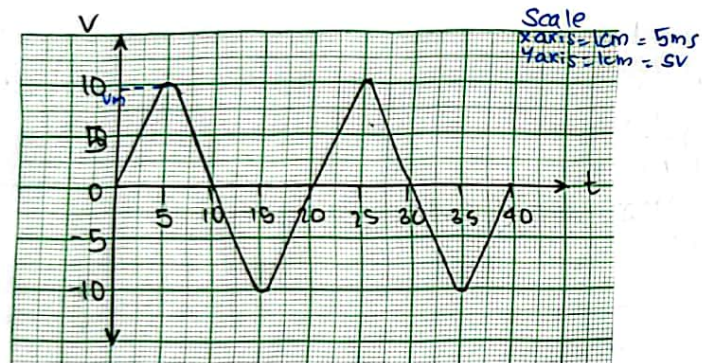
The full wave rectifier circuit of two diodes connected to a single load resistor (R_L) with each diode taking it in turn to supply current to the load. When point A of the transformer is positive with respect to point C, diode D_1 conducts in the forward direction as indicated by the arrow.

When point B is positive (in the negative half of the cycle) with respect to point C, diode D_2 conducts in the forward direction and the current flowing through resistor R_L is in the same direction for both half cycles.

Filter

In this arrangement a high value capacitor is placed directly across the output terminals. During the positive half cycle will be charged to the peak voltage. Once it is charged the diode will get reversed biased since the voltage at the positive terminal is

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less than peak voltage on the capacitor. This will stop the conduction of the diode and the capacitor will discharge through the load resistor.

PROCEDURE

1. Set up the circuit on the bread board
2. Switch on the supply
3. Note down the waveform of secondary voltage and output voltage using CRO
4. Plot the input and output wave forms in graph.

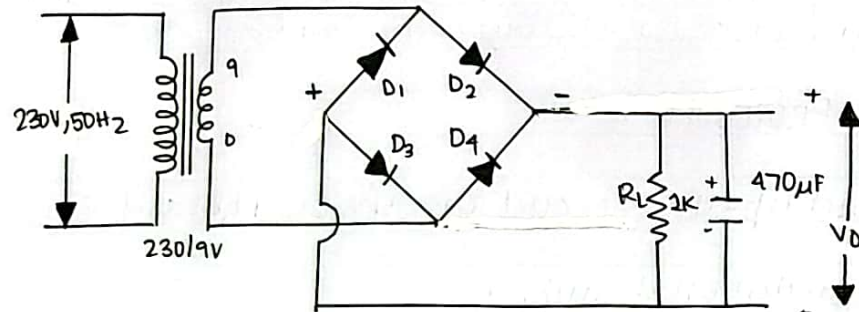
RESULT

A half and full wave rectifier was created with and without filter and was studied and the load regulation graph was plotted.

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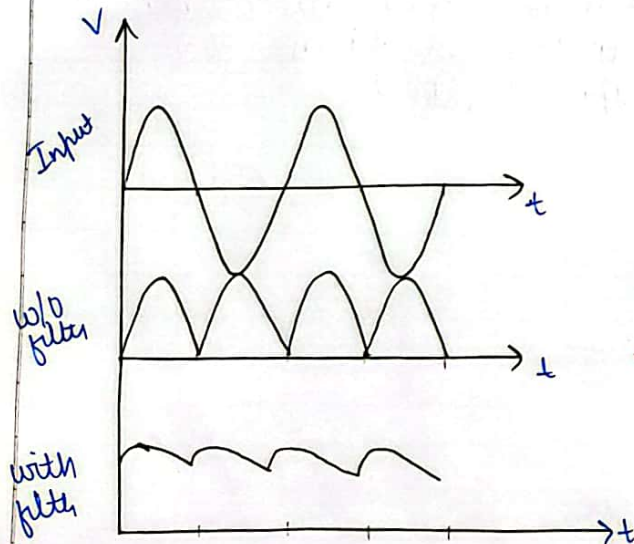
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CIRCUIT DIAGRAM



OBSERVATION

Waveform



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BRIDGE RECTIFIER

AIM

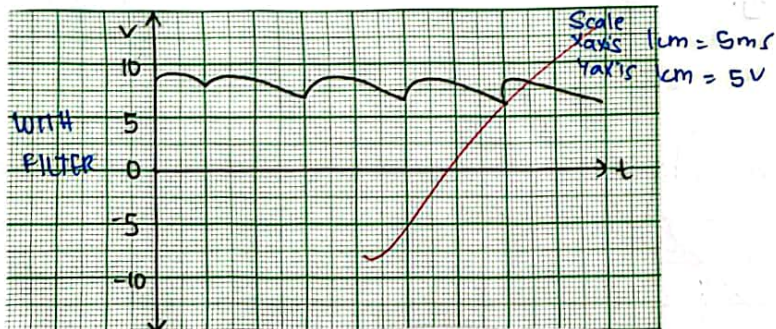
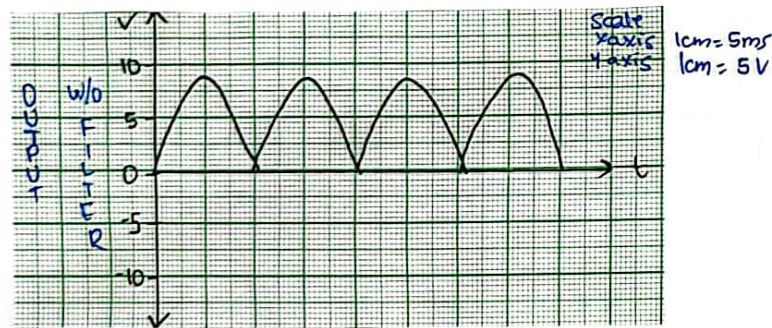
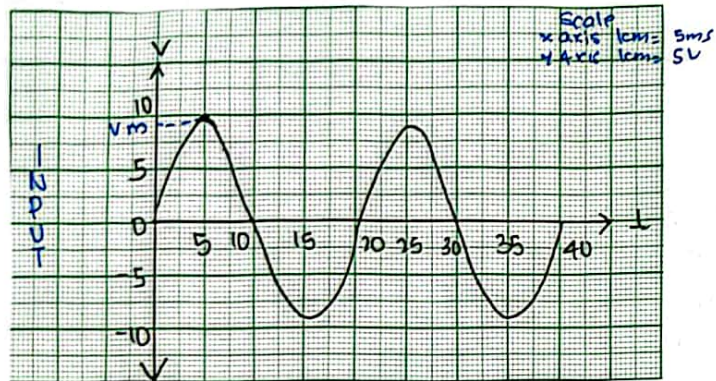
To study the performance of Bridge Rectifier w/o filter

- 1) with filter
- 2) without filter

MATERIALS AND EQUIPMENT REQUIRED

S.No	Item	Specification	Quantity
1.	Transformer	9V, 500mA	1
2.	Diode	1N4007	4
3.	Resistor	1KΩ	1
4.	Capacitor	470μF, 63V	1
5.	CRD		
6.	Breadboard		
7.	Connecting wire		

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THEORY

The bridge rectifier is a circuit which converts an AC voltage to DC voltage using both half cycles of the input AC voltage. The circuit has four diodes connected to form a bridge. The AC input voltage is applied to the diagonally opposite ends of the bridge. The load resistance is connected between the other two ends of the bridge. For the positive half cycle of the input AC voltage diodes D2 and D3 conduct, whereas diodes D1 and D4 remain in the OFF state. The conducting diodes will be in series with the load resistance R_L and hence the load current flows through R_L . For the negative half cycle of the input AC voltage, diodes D1 and D4 conduct whereas D2 and D3 remain OFF. The conducting diodes D1 and D4 will be in series with the load resistance R_L and hence the current flows through R_L in the same direction as in the previous half cycle. Thus a bi-directional wave is converted into a unidirectional wave.

PROCEDURE

1. Set up the circuit on the breadboard without filter as shown in figure.
2. Switch on the supply.
3. Note down the waveforms of the transformer.

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secondary voltage and output voltage

1. Connect the capacitor filter and observe the waveform.

RESULT

~~A bridge rectifier was created with and without filter and was studied and the load regulation graph was plotted.~~

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